

Smart Fisheye Camera Monitoring System

Smartphone-Based Edge Processing with Remote Camera Control

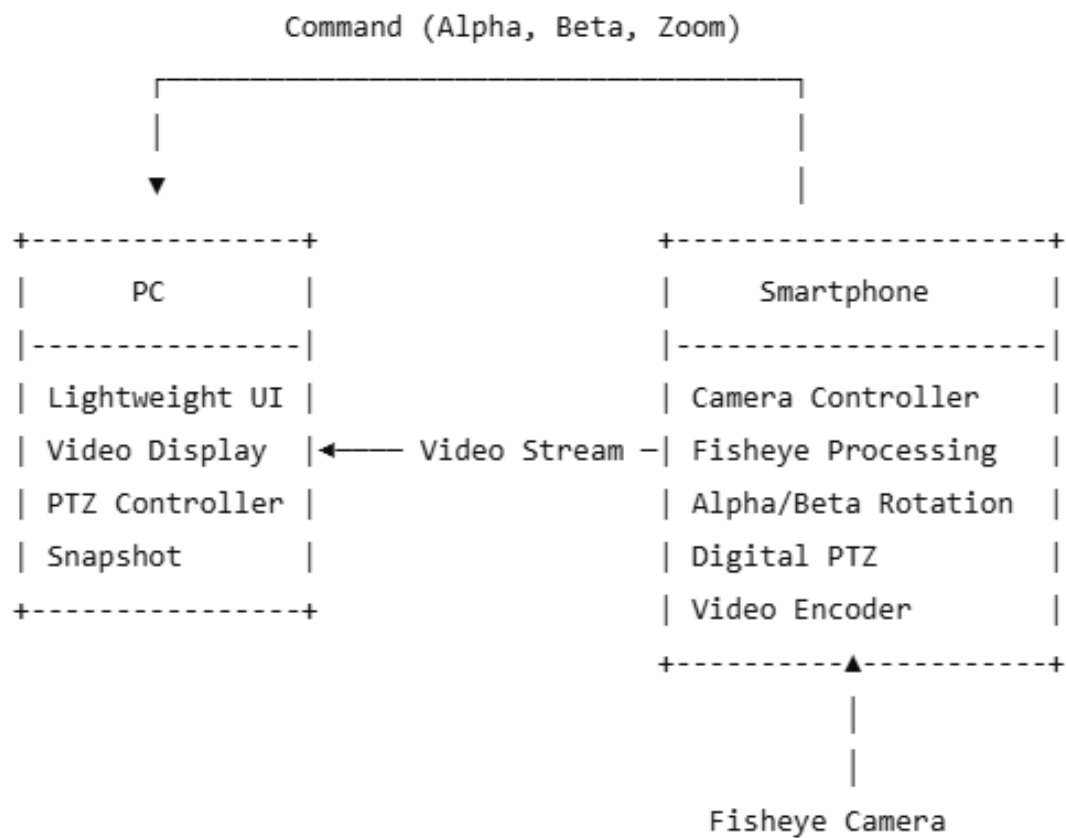
Executive Summary

This project proposes a distributed monitoring system where a smartphone serves as the primary image processing unit and a PC function as a lightweight monitoring and control terminal.

The smartphone receives the raw fisheye camera stream, performs all image processing—including fisheye transformation, digital PTZ, and view generation—and streams the processed video to the PC. Rather than processing video locally, the PC sends control commands to the smartphone, allowing operators to interactively change the viewing direction by adjusting the virtual camera's **alpha** and **beta** angles.

This architecture significantly reduces PC hardware requirements while maintaining a responsive and interactive monitoring experience.

System Architecture



Operating Principle

1. The fisheye camera continuously captures the raw image.
2. The smartphone receives the raw camera data.
3. The smartphone performs all image transformations in real time.
4. The processed view is encoded and streamed to the PC.
5. The operator uses the PC interface to change the viewing direction.
6. The PC transmits control commands (Alpha, Beta, Zoom, Preset) to the smartphone.
7. The smartphone updates the virtual camera orientation and immediately generates the new view.
8. The updated image is streamed back to the PC with minimal latency.

Command Interface

The PC communicates only through lightweight control commands rather than transferring image processing workloads.

Example commands:

- Set Alpha Angle
- Set Beta Angle
- Digital Zoom In/Out
- Reset View
- Panorama View
- Preset Positions
- Auto Rotation
- Image Quality Settings
- Frame Rate Adjustment

Example JSON command:

```
{  
  "command": "set_view",  
  "alpha": 35.0,  
  "beta": -20.0,  
  "zoom": 1.5  
}
```

Smartphone Responsibilities

- Capture fisheye images/videos
- Perform fisheye correction
- Generate perspective views
- Apply Alpha/Beta rotation
- Digital PTZ
- Image enhancement
- Hardware video encoding (H.264 recommended)
- Receive remote control commands
- Stream processed video

PC Responsibilities

The PC is intentionally lightweight and performs only:

- Receive live video
- Decode video stream
- Display video
- Send Alpha/Beta control commands
- Zoom control
- Preset management
- Snapshot
- Recording (optional)

No image transformation is executed on the PC.

Communication Protocol

Two communication channels are recommended:

Video Channel

- Protocol: RTP/RTSP or WebRTC
- Codec: Hardware H.264
- Purpose: Live video streaming

Control Channel

- Protocol: MQTT, WebSocket, or TCP
- Payload: JSON commands
- Purpose: Alpha/Beta control, zoom, presets, and system configuration

Separating video and control channels improves responsiveness, simplifies system maintenance, and enables future expansion.

Key Advantages

- Extremely lightweight PC application
- Low CPU and GPU utilisation on the PC
- High responsiveness through lightweight control commands
- Efficient use of smartphone hardware acceleration
- Lower network bandwidth compared to transmitting raw fisheye data
- Scalable architecture supporting multiple monitoring clients
- Easy integration with future AI features running on the smartphone

Future Enhancements

- Multi-view generation from a single fisheye image
- AI object detection and tracking
- Automatic target following
- Preset patrol modes
- Cloud monitoring
- Remote firmware updates
- Multi-user access control
- Mobile-to-mobile monitoring